ITA03-MP Assignment 3

MOBILE COMPUTING

ASSIGNMENT 3 – CO3

Mode: Pair Work – 2 Students per Group
Duration: 1 Week
Total Marks: 50

Pair-Work Principle

Both students are responsible for understanding the complete assignment.

However, each question contains clearly identified individual responsibilities so that the contribution and understanding of each student can be independently verified.

Student A and Student B must not divide the assignment question-wise. Both students must participate in every question.

The individual responsibilities intentionally alternate across questions so that both students demonstrate evidence of:

Concept Understanding → Network Analysis → Protocol Design → Implementation/Simulation → Evaluation

Q1 – Wireless Network Architecture and Mobility Analysis 6 Marks

Part A – Introduction to Wireless Networks

Shared Group Work

Together:

- Understand the characteristics and components of wireless networks.
- Identify mobile nodes, wireless access points/base stations and communication links.
- Develop a suitable wireless network architecture.
- Identify bandwidth, connectivity and mobility constraints.
- analyse communication between mobile users and network infrastructure.
- Implement or simulate a small wireless communication scenario and test the communication flow.

Student A – Network Architecture Analysis

Student A must individually:

- Draw and represent the wireless network architecture.
- Identify the major network components and their roles.
- analyse wireless communication characteristics.
- Analyse the effect of limited bandwidth and wireless connectivity.
- analyse how mobile nodes communicate through the wireless infrastructure.

Student B – Mobility and Performance Analysis

Student B must individually:

- Design representative mobility scenarios.
- Analyse handoff and connectivity interruption.
- Study packet loss and transmission delay.
- Analyse the effect of changing network conditions.
- Compare communication behaviour under different mobility conditions.

Part B – Wireless Mobility Scenario

Shared Group Work

Together, develop and analyse a scenario in which a mobile user moves between wireless coverage areas.

Student A – Analytical Investigation

Student A must:

- Define successful communication and interrupted communication cases.
- Identify the sequence of mobility events.
- Record communication delay and packet delivery behaviour.
- Investigate increasing mobility conditions.
- Present the analytical evidence.

Student B – Mobility Logic

Student B must:

- Formulate the mobility decision sequence.
- analyse when a mobile node changes its point of attachment.

- Identify which communication conditions are affected.
- analyse the termination or completion condition for the scenario.
- Relate repeated mobility to communication efficiency.

Individual Verification

Faculty may ask Student A to analyse the wireless architecture and Student B to analyse the mobility scenario. Therefore, primary responsibility does not mean exclusive knowledge.

Q2 – Mobile IP and Mobility Management **7 Marks**

Shared Group Work

The pair must:

- Study Mobile IP architecture and terminology.
- Identify Mobile Node, Home Agent, Foreign Agent and Correspondent Node.
- analyse agent discovery and registration.
- Analyse packet delivery to a mobile node.
- analyse tunnelling and mobility support.
- Conduct a small simulation or worked example of Mobile IP communication.

Student A – Mobile IP Operation

Student A is individually responsible for:

- analyseing the role of the Mobile Node.
- analyseing the role of the Home Agent.
- analyseing the role of the Foreign Agent.
- analyseing the role of the Correspondent Node.
- Describing registration and location updating.
- analyseing the packet-delivery and tunnelling process.

Student B – Mobile IP Challenge Analysis

Student B is individually responsible for:

- Identifying Mobile IP challenges.
- Analysing registration delay.
- Analysing tunnelling overhead.
- analyseing triangular routing.

- Analysing packet loss during mobility.
- interpreting security and scalability considerations.

Joint Design Question

Together answer:

If a mobile user frequently changes its network of attachment during an active communication session, what problems can occur in Mobile IP and how can they be reduced?

Evidence Expected

- Mobile IP architecture diagram.
- Registration/packet-flow sequence.
- Identification of mobility-related delays and overhead.
- Reasoned justification of the proposed improvement.

Cross-Understanding Requirement

Student A must be able to analyse the major Mobile IP challenges.

Student B must be able to analyse the complete Mobile IP registration and packet-delivery process.

Both students must be able to defend the answer.

Q3 – Cellular Networks versus Ad Hoc Wireless Networks 7 Marks

Shared Group Work

Together:

- Study cellular wireless network architecture.
- Study ad hoc wireless network architecture.
- Identify infrastructure-based and infrastructure-less communication.
- Compare mobility and topology characteristics.
- Compare routing requirements.
- Develop a technical comparison diagram and table.

Student A – Cellular Network Investigation

Student A must individually:

- Represent cellular network architecture.
- analyse the role of base stations and infrastructure.
- Analyse centralized communication.
- analyse handoff between cells.
- Analyse scalability and coverage.
- Identify suitable cellular applications.

Student B – Ad Hoc Network Investigation

Student B must individually:

- Represent ad hoc network architecture.
- analyse communication between mobile nodes.
- Analyse dynamic topology.
- analyse multi-hop communication.
- Analyse the absence of fixed infrastructure.
- Identify suitable ad hoc applications.

Comparative Analysis

Parameter	**Cellular Network**	**Ad Hoc Network**
Infrastructure		
Communication		
Routing		
Mobility		
Topology		
Scalability		
Handoff		
Failure Handling		
Applications		

Joint Engineering Decision

Together determine:

For an emergency/disaster environment where fixed infrastructure may be unavailable, which network architecture would you select and why?

Cross-Defence

Student A must analyse why cellular networks depend on infrastructure.

Student B must analyse how ad hoc networks continue communication without fixed infrastructure.

Q4 – Applications of Mobile IP and Routing Requirements 7 Marks

Shared Group Work

Together:

- Identify real-world applications requiring mobility support.
- Select one application scenario such as emergency communication, mobile healthcare or intelligent transportation.
- Identify the mobile entities and communication requirements.
- Apply Mobile IP concepts to the selected scenario.
- Identify the routing requirements created by mobility.
- Evaluate the suitability of the proposed solution.

Student B – Application Architecture

Student B must individually:

- Select and describe an appropriate Mobile IP application.
- Identify mobile nodes and communication endpoints.
- Design the application network architecture.
- analyse how Mobile IP supports user mobility.
- Identify required network components.
- analyse the communication flow.

Student A – Performance and Requirement Analysis

Student A must individually:

- Analyse communication delay.
- Analyse packet delivery and connectivity.
- Identify mobility requirements.
- Analyse interruption during movement.
- Identify security and reliability requirements.

- Evaluate Mobile IP suitability.

Q5 – Proactive Routing Protocols in Mobile Networks **6 Marks**

Shared Group Work

Together:

- Study proactive routing protocols.
- Understand continuous routing-table maintenance.
- Select a suitable protocol such as DSDV.
- Develop a mobile-node topology.
- Analyse routing-table updates.
- Evaluate routing behaviour under mobility.

Student A – Protocol Design

Student A must individually:

- analyse the principle of proactive routing.
- analyse DSDV operation.
- analyse routing-table maintenance.
- analyse sequence numbers and route updates.
- Draw a sample routing topology.
- Demonstrate route information before and after a topology change.

Student B – Performance Investigation

Student B must individually:

- Analyse routing-control overhead.
- Analyse route availability.
- Study convergence behaviour.
- Analyse control-message transmission.
- Compare behaviour for different network sizes.
- Evaluate delay and packet delivery.

Q6 – Reactive Routing and Protocol Comparison **7 Marks**

Student A – Reactive Routing Investigation

Student A takes primary responsibility for reactive routing.

- analyse the concept of reactive routing.
- Study AODV.
- analyse Route Request (RREQ).
- analyse Route Reply (RREP).
- analyse Route Error (RERR).
- Analyse route discovery and route maintenance.
- analyse the effect of mobility on route discovery.

Student B – Comparative Verification

Student B must provide a comparative analysis of AODV and DSDV.

- Compare route discovery mechanisms.
- Compare routing-table requirements.
- Analyse control overhead.
- Compare initial communication delay.
- Analyse mobility handling.
- Identify suitable application scenarios.
- Justify the selection of a protocol for a given network condition.

Joint Question

For a highly dynamic mobile network with frequent topology changes, would you select DSDV or AODV? Justify your decision using routing overhead, delay and mobility considerations.

DAY-4 DESIGN CHALLENGE

Q7 – Design a Mobile Emergency Communication Network 10 Marks

The Day-4 problem must be solved collaboratively.

Neither student may work on this question independently and simply combine the answers later.

Scenario

During a natural disaster, ambulances, police vehicles, rescue teams and medical personnel move continuously across different geographical areas. Fixed communication infrastructure may become unavailable, while mobile users must exchange critical information with minimum delay.

Stage 1 – Independent Design Proposal

Before interpreting the solution with each other:

Student A

Independently propose:

- A suitable wireless network architecture.
- A mobility-management strategy.
- The role of Mobile IP.
- A possible communication model.

Student B

Independently propose:

- A suitable routing strategy.
- Proactive/reactive protocol selection.
- Route maintenance approach.
- Failure-handling mechanism.
- Performance parameters.

Each student's initial proposal must be retained in the submission.

Stage 2 – Compare the Two Designs

The pair must compare where the designs agree, where they differ, strengths and weaknesses, Mobile IP requirements, routing suitability, expected overhead, expected delay and handling of node mobility.

Stage 3 – Joint Final Design

Using the strongest ideas from the interpretion, jointly develop the final mobile emergency communication solution.

- Network architecture showing mobile nodes, wireless links and routing infrastructure.
- Mobile IP components: Mobile Node, Home Agent, Foreign Agent and Correspondent Node.
- Registration and tunnelling process.
- Selected routing protocol and justification.
- Failure handling for node/link failure and network disconnection.
- Performance evaluation using delay, packet delivery and routing overhead.

Stage 4 – Divide the Validation Responsibilities

Student A – Analytical Validation

- Analyse Mobile IP overhead.
- Analyse mobility-related delay.
- Evaluate packet delivery.
- Analyse communication interruption.
- Justify the selected mobility-management mechanism.

Student B – Experimental Validation

- Design network test cases.
- Test different numbers of mobile nodes.
- Test different mobility conditions.
- Compare proactive and reactive routing behaviour.
- Record delay and packet-delivery results.
- Present experimental evidence.

Stage 5 – Cross-Defence

Student A must analyse: Why was the selected routing protocol appropriate for the emergency mobile network?

Student B must analyse: How does Mobile IP maintain communication when a mobile node changes its network location?

Stage 6 – Joint Evaluation

- Why the proposed network architecture is suitable.
- Why Mobile IP is required or not required.
- Why the selected routing protocol is appropriate.
- What network overhead is introduced.
- How mobility affects performance.
- What happens when connectivity is lost.
- Under what conditions the proposed solution may not be appropriate.

Conclude: What characteristics should a network designer analyse before selecting Mobile IP and a routing protocol for a highly mobile wireless environment?

CONTRIBUTION RECORD

Question	**Student A – Primary Contribution**	**Student B – Primary Contribution**	**Joint Work**

Q1	Network Architecture	Mobility Analysis	Wireless Network Analysis
Q2	Mobile IP Operation	Mobile IP Challenges	Mobile IP Architecture
Q3	Cellular Network	Ad Hoc Network	Comparative Analysis
Q4	Performance Analysis	Application Architecture	Application Evaluation
Q5	Protocol Design	Performance Investigation	Proactive Routing
Q6	Reactive Routing	DSDV–AODV Comparison	Routing Evaluation
Q7	Analytical Validation	Experimental Validation	Final Network Design

INDIVIDUAL VERIFICATION

Faculty may select any question, including a component for which the student was not designated as the primary contributor.

- analyse wireless network architecture.
- analyse Mobile IP operation and registration.
- analyse mobility challenges.
- Compare cellular and ad hoc wireless networks.
- analyse applications of Mobile IP.
- analyse proactive routing.
- analyse reactive routing.
- Compare DSDV and AODV.
- Justify protocol selection.
- Interpret experimental results.
- Justify network design decisions.
- Respond to a modified network condition.

A student should not answer: “My partner did this section.”

Understanding of the complete submitted work is a requirement of the assignment.

NOTE: RECOMMENDED FACULTY VERIFICATION MODEL (VIVA)

For each student, conduct a brief 4–5 minute verification.

Question 1 – Primary responsibility

Ask one technical question from the component for which the student had primary responsibility.

Question 2 – Partner's responsibility

Ask one question from a component primarily completed by the partner.

Question 3 – Transfer question

Modify a condition and ask the student to reason about the consequence.

- If a mobile node moves from one network to another during an active session, what happens to its IP connectivity and how does Mobile IP address the problem?
- If the number of mobile nodes increases significantly, how would AODV and DSDV differ in routing overhead?
- If the network topology changes very frequently, which routing approach would you select and why?
- If the fixed infrastructure becomes unavailable during a disaster, how would an ad hoc network support communication?

This verification establishes individual attainment without requiring a second complete individual assignment.

SIMATS ENGINEERING

MOBILE COMPUTING

STANDARD FOR ALGORITHMIC PSEUDOCODE

Applicable to: Mobile COmputing Assignments and Exercises

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1. What is Mobile Computing Pseudocode?

Mobile Computing pseudocode is a clear, precise and language-independent representation of a mobile networking algorithm. It describes mobility management, packet forwarding, route discovery, route maintenance, handoff, and routing decisions without depending on Java, Python, C, C++, Android, or simulator-specific syntax.

A good Mobile Computing algorithm must make the network state and communication logic visible. It should show what happens when a mobile node moves, when a route exists or fails, when a registration is required, and when a packet must be forwarded through a tunnel or discovered route.

Remember

- Pseudocode is not English prose.
- Pseudocode is not programming code.
- Pseudocode is an implementation-independent specification of mobile-network logic.
- Every important networking decision must be explicit.
- Abstraction is permitted; ambiguity is not.

2. Standard Structure

Use the following structure for a complete Mobile Computing algorithm.

ALGORITHM AlgorithmName(parameters)

INPUT:

Clearly describe nodes, packets, addresses, routes or network state

OUTPUT:

Clearly describe the expected mobility/routing result

BEGIN

INITIALISE required variables/data structures

Detect or receive relevant network information

Execute mobility/routing decisions

UPDATE network state

RETURN result

END

Example

ALGORITHM CheckMobileNodeLocation(MN, HomeNetwork, CurrentNetwork)

INPUT:

MN – mobile node

HomeNetwork – node's registered home network

CurrentNetwork – network where MN is currently attached

OUTPUT:

Mobility status

BEGIN

IF CurrentNetwork = HomeNetwork THEN

status ← HOME_NETWORK

ELSE

status ← VISITED_NETWORK

END IF

RETURN status

END

3. Standard Keywords and Notation

PURPOSE	USE
Declare algorithm	ALGORITHM
Specify input	INPUT:
Specify output	OUTPUT:
Start algorithm	BEGIN
End algorithm	END
Assignment	←
Condition	IF ... THEN
Alternative	ELSE
Multiple conditions	ELSE IF
Loop	FOR ... DO / WHILE ... DO
Return result	RETURN
Boolean operations	AND, OR, NOT
Packet operation	FORWARD / DELIVER / DROP
Route operation	ADD ROUTE / REMOVE ROUTE / UPDATE ROUTE
Message operation	SEND / RECEIVE
Timer	START TIMER / EXPIRE TIMER

Use explicit state names such as HomeAddress, CareOfAddress, NextHop, SequenceNumber, HopCount, RouteState, LinkState and RegistrationStatus.

4. Wireless Network Representation

Represent a mobile network as a graph G = (V, E), where V is the set of mobile nodes and infrastructure nodes, and E represents currently usable communication links.

NETWORK G = (V, E)

V = {mobile nodes, base stations, access points, gateways}

E = {wireless links currently available}

link(u,v) = communication link between u and v

cost(u,v) = communication cost of link (u,v)

status(u,v) ∈ {UP, DOWN}

Network State

STATE:

Location[MN] = current network/location of mobile node MN

NextHop[D] = next hop toward destination D

RouteState[D] ∈ {VALID, INVALID, UNKNOWN}

SequenceNumber[D] = latest known destination sequence number

HopCount[D] = number of hops in current route

LastUpdate[D] = time of most recent route information

5. Wireless Network Discovery

Wireless nodes must determine whether they are connected to a suitable network and whether neighboring nodes are reachable. The discovery procedure is especially important in dynamic environments.

ALGORITHM DiscoverWirelessNeighbors(MN, TransmissionRange)

INPUT:

MN – mobile node

TransmissionRange – maximum discovery range

OUTPUT:

NeighborSet

BEGIN

NeighborSet ← EMPTY_SET

BROADCAST HELLO from MN

WAIT for neighbor responses

FOR each Response received DO

IF SignalStrength(Response) > MIN_SIGNAL THEN

IF Distance(MN, Response.Node) ≤ TransmissionRange THEN

ADD Response.Node to NeighborSet

END IF

END IF

END FOR

RETURN NeighborSet

END

6. Mobile IP – Basic Operation

Mobile IP allows a mobile node to retain a permanent home address while changing its point of network attachment. The algorithm must maintain a binding between the permanent home address and the current care-of address.

Mobile IP Entities

- Mobile Node (MN)
- Home Network
- Home Agent (HA)
- Foreign Network / Visited Network
- Foreign Agent (FA) in traditional Mobile IPv4 terminology
- Care-of Address (CoA)
- Correspondent Node (CN)

ALGORITHM MobileIPLocationUpdate(MN, HA, CurrentNetwork)

INPUT:

MN – mobile node

HA – home agent

CurrentNetwork – visited network

OUTPUT:

Registered care-of address

BEGIN

IF CurrentNetwork = MN.HomeNetwork THEN

CoA ← MN.HomeAddress

RegistrationStatus ← HOME

ELSE

CoA ← ObtainCareOfAddress(CurrentNetwork)

SEND RegistrationRequest(MN.HomeAddress, CoA) TO HA

WAIT FOR RegistrationReply

IF RegistrationReply = ACCEPTED THEN

RegistrationStatus ← REGISTERED

ELSE

RegistrationStatus ← FAILED

END IF

END IF

RETURN CoA, RegistrationStatus

END

7. Mobile IP Agent Discovery

ALGORITHM MobileIPAgentDiscovery(MN)

INPUT:

MN – mobile node

OUTPUT:

Detected mobility agent or network information

BEGIN

LISTEN for AgentAdvertisement

IF Advertisement received THEN

Extract NetworkAddress

Extract AgentType

IF AgentType = HOME_AGENT THEN

Set HomeAgentAddress

ELSE IF AgentType = FOREIGN_AGENT THEN

Set ForeignAgentAddress

END IF

RETURN AgentType, NetworkAddress

ELSE

SEND AgentSolicitation

WAIT for Advertisement

IF Advertisement received THEN

RETURN detected agent information

ELSE

RETURN AGENT_NOT_FOUND

END IF

END IF

END

8. Mobile IP Registration

ALGORITHM RegisterMobileNode(MN, HA, CoA)

INPUT:

MN – mobile node

HA – home agent

CoA – current care-of address

OUTPUT:

Registration result

BEGIN

Create RegistrationRequest

Request.HomeAddress ← MN.HomeAddress

Request.CareOfAddress ← CoA

Request.Lifetime ← RequestedLifetime

Request.Identifier ← GenerateRegistrationIdentifier()

Authenticate Request

SEND Request TO HA

WAIT FOR RegistrationReply

IF Reply.Status = ACCEPTED THEN

Binding[MN.HomeAddress] ← CoA

RegistrationExpiry ← CurrentTime + Reply.Lifetime

RETURN REGISTRATION_SUCCESS

ELSE

RETURN REGISTRATION_FAILURE

END IF

END

9. Mobile IP Packet Tunnelling

The home agent can forward packets toward the mobile node's current location by tunnelling them to the registered care-of address.

ALGORITHM TunnelPacketToMobileNode(Packet, DestinationHomeAddress)

INPUT:

Packet – original IP packet

DestinationHomeAddress – mobile node home address

OUTPUT:

Packet delivered through tunnel or DROP

BEGIN

IF Binding[DestinationHomeAddress] does not exist THEN

DROP Packet

RETURN DELIVERY_FAILURE

END IF

CoA ← Binding[DestinationHomeAddress]

EncapsulatedPacket ← ENCAPSULATE(Packet, CoA)

FORWARD EncapsulatedPacket TO CoA

IF TunnelDeliverySuccessful THEN

Decapsulate at tunnel endpoint

DELIVER original Packet TO MobileNode

RETURN DELIVERY_SUCCESS

ELSE

RETURN DELIVERY_FAILURE

END IF

END

10. Mobile IP Challenges

CHALLENGE	**ALGORITHMIC IMPLICATION**
Handoff delay	Detect movement and update location quickly
Registration overhead	Limit unnecessary signalling
Packet loss	Buffer, retransmit or reroute when appropriate

Triangular routing	Evaluate indirect path and forwarding delay
Security	Authenticate registration and protect mobility messages
Scalability	Avoid excessive control traffic
Wireless errors	Detect link failure and retry or select alternate path
Changing bandwidth	Adapt routing and transmission decisions

Challenge Detection Algorithm

ALGORITHM DetectMobileIPChallenge(MN, Route, Registration)

INPUT:

MN, current route, registration state

OUTPUT:

Identified challenge

BEGIN

IF RegistrationStatus = FAILED THEN

RETURN REGISTRATION_PROBLEM

ELSE IF HandoffInProgress = TRUE THEN

RETURN HANDOFF_DELAY

ELSE IF PacketLossRate > MAX_LOSS THEN

RETURN PACKET_LOSS

ELSE IF RouteDelay > MAX_DELAY THEN

RETURN ROUTING_DELAY

ELSE IF SecurityValidation = FAILED THEN

RETURN SECURITY_PROBLEM

ELSE

RETURN NO_MAJOR_PROBLEM

END IF

END

11. Handoff Management

ALGORITHM PerformHandoff(MN, OldNetwork, NewNetwork)

INPUT:

MN – mobile node

OldNetwork – previous point of attachment

NewNetwork – target network

OUTPUT:

Handoff status

BEGIN

Detect NewNetwork

Measure signal quality

IF SignalQuality(NewNetwork) < MIN_SIGNAL THEN

RETURN HANDOFF_REJECTED

END IF

Obtain new network address or care-of address

Pause non-critical transmissions if required

UPDATE mobility registration

UPDATE routing information

IF RegistrationStatus = SUCCESS THEN

SWITCH active attachment TO NewNetwork

Resume data transmission

RETURN HANDOFF_SUCCESS

ELSE

Restore OldNetwork if available

RETURN HANDOFF_FAILURE

END IF

END

12. Cellular vs Ad Hoc Wireless Networks

PARAMETER	CELLULAR NETWORK	AD HOC NETWORK
Infrastructure	Base stations / access infrastructure	No fixed infrastructure required
Topology	Planned cells	Dynamic topology
Routing	Infrastructure-assisted	Distributed multi-hop routing
Node role	Mobile device mainly communicates through infrastructure	A node can act as host and router
Mobility support	Cellular mobility management	Distributed route discovery/maintenance
Deployment	Requires pre-existing infrastructure	Can be rapidly deployed
Typical environment	Public cellular service	Emergency, military, temporary or field networks
Failure behaviour	Infrastructure failure can affect service	Alternate peer routes may be possible

13. Selecting Cellular or Ad Hoc Architecture

ALGORITHM SelectWirelessArchitecture(Scenario)

INPUT:

InfrastructureAvailability

NodeMobility

DeploymentTime

CoverageRequirement

MultiHopRequirement

OUTPUT:

Recommended architecture

BEGIN

IF InfrastructureAvailability = AVAILABLE AND

CoverageRequirement = LARGE THEN

Architecture ← CELLULAR

ELSE IF DeploymentTime = IMMEDIATE AND

InfrastructureAvailability = UNAVAILABLE THEN

Architecture ← AD_HOC

ELSE IF MultiHopRequirement = REQUIRED THEN

Architecture ← AD_HOC

ELSE

Architecture ← CELLULAR

END IF

RETURN Architecture

END

14. Applications of Mobile IP

- Mobile employees accessing organizational services while moving between networks.
- Continuous Internet access across different points of attachment.
- Emergency response and disaster recovery communication.
- Mobile healthcare and ambulance communication.
- Transportation and fleet monitoring.
- Field engineering and remote maintenance.
- Temporary communication networks.
- Location-independent access to network applications.

ALGORITHM SelectMobileIPApplication(Requirement)

INPUT:

Requirement – mobility and communication requirement

OUTPUT:

Suitable application category

BEGIN

IF Requirement requires continuous Internet access while moving THEN

RETURN MOBILE_INTERNET_ACCESS

ELSE IF Requirement involves emergency vehicle mobility THEN

RETURN EMERGENCY_COMMUNICATION

ELSE IF Requirement requires remote patient/device connectivity THEN

RETURN MOBILE_HEALTHCARE

ELSE IF Requirement involves moving vehicles or fleets THEN

RETURN TRANSPORTATION

ELSE

RETURN GENERAL_MOBILE_ACCESS

END IF

END

15. Routing Protocols in Mobile Networks

Routing protocols determine how packets travel through changing mobile topologies. For this unit, routing is studied using the two major approaches: proactive and reactive.

Routing State

STATE:

RouteTable[Destination] = {NextHop, HopCount, SequenceNumber, State}

State $\in$ {VALID, INVALID, UNKNOWN}

NeighborTable = currently reachable neighbors

RouteRequestID = identifier for route discovery

RouteLifetime = validity period

16. Proactive Routing

A proactive protocol maintains route information before a packet transmission request occurs. This can reduce route-discovery delay but may create periodic control overhead.

ALGORITHM ProactiveRouteMaintenance(Node)

INPUT:

Node – mobile node

OUTPUT:

Updated routing table

BEGIN

INITIALISE RouteTable

PERIODICALLY DO

Broadcast current routing information

RECEIVE routing updates from neighbors

FOR each Update DO

Validate Update

FOR each Destination in Update DO

Calculate NewMetric

IF NewMetric < CurrentMetric(Destination) OR

Update.SequenceNumber > CurrentSequenceNumber(Destination) THEN

UPDATE RouteTable[Destination]

Set NextHop and HopCount

END IF

END FOR

END FOR

Remove expired or invalid entries

END PERIODICALLY

RETURN RouteTable

END

17. Proactive Routing – DSDV-Style Sequence-Number Update

A destination-sequenced distance-vector approach uses route metrics and sequence numbers to maintain fresh routes and reduce routing loops.

ALGORITHM UpdateDSDVRoute(Node, ReceivedRoute)

INPUT:

Node – receiving node

ReceivedRoute = {Destination, NextHop, HopCount, SequenceNumber}

OUTPUT:

Updated routing table

BEGIN

Destination ← ReceivedRoute.Destination

NewHopCount ← ReceivedRoute.HopCount + 1

NewSequence ← ReceivedRoute.SequenceNumber

IF Destination not in RouteTable THEN

ADD ReceivedRoute with NextHop = Node.Neighbor

ELSE IF NewSequence > RouteTable[Destination].SequenceNumber THEN

REPLACE RouteTable[Destination]

ELSE IF NewSequence = RouteTable[Destination].SequenceNumber AND

NewHopCount < RouteTable[Destination].HopCount THEN

REPLACE RouteTable[Destination]

END IF

RETURN RouteTable

END

18. Reactive Routing

A reactive protocol discovers a route only when a source needs to send data and no valid route is available. Route discovery and route maintenance are therefore demand-driven.

ALGORITHM ReactiveRouteDiscovery(Source, Destination)

INPUT:

Source – requesting node

Destination – required destination

OUTPUT:

Route or ROUTE_NOT_FOUND

BEGIN

IF RouteTable[Destination].State = VALID THEN

RETURN RouteTable[Destination]

END IF

RouteRequestID ← RouteRequestID + 1

Create RouteRequest(Source, Destination, RouteRequestID)

Broadcast RouteRequest to neighbors

WAIT for RouteReply or RouteFailure

IF RouteReply received THEN

Establish reverse path

Establish forward path

Mark RouteTable[Destination] as VALID

RETURN RouteTable[Destination]

ELSE

RETURN ROUTE_NOT_FOUND

END IF

END

19. Reactive Routing – AODV-Style Route Request

ALGORITHM ProcessRouteRequest(Node, Request)

INPUT:

Node – current receiving node

Request = {Source, Destination, RequestID, HopCount}

OUTPUT:

Route discovery action

BEGIN

IF Request already processed THEN

DISCARD Request

RETURN

END IF

Record reverse route toward Request.Source

Mark Request as processed

IF Node = Request.Destination THEN

Create RouteReply

SEND RouteReply toward Source

RETURN

END IF

IF ValidRouteTo(Request.Destination) THEN

Create RouteReply using valid route

SEND RouteReply toward Source

ELSE

Request.HopCount ← Request.HopCount + 1

FOR each Neighbor in NeighborTable DO

FORWARD Request TO Neighbor

END FOR

END IF

END

20. Reactive Route Maintenance

ALGORITHM MaintainReactiveRoute(Source, Destination)

INPUT:

Source and Destination

OUTPUT:

Route status

BEGIN

MONITOR active route

IF LinkFailureDetected THEN

Mark failed link as INVALID

IF AlternateValidRoute exists THEN

Select alternate route

RETURN ALTERNATE_ROUTE_USED

ELSE

Generate RouteError

SEND RouteError toward affected source

IF Source still requires communication THEN

Initiate new RouteDiscovery

ELSE

RETURN ROUTE_TERMINATED

END IF

END IF

END IF

RETURN ROUTE_ACTIVE

END

21. Proactive vs Reactive Routing

FEATURE	**PROACTIVE**	**REACTIVE**
Route availability	Maintained before data transmission	Obtained when required
Route discovery delay	Low	Present when route is unknown
Control overhead	Periodic/continuous	Demand-driven
Stale information	Possible	Route is generally discovered when needed
Frequent traffic	Often suitable	May repeatedly use discovered routes
Infrequent traffic	May waste control traffic	Often efficient
Topology changes	Routing tables updated periodically	Route repair/discovery triggered
Typical example	DSDV-style	AODV-style

22. Proactive or Reactive – Decision Algorithm

ALGORITHM SelectRoutingApproach(TrafficRate, MobilityRate, DelayRequirement, Bandwidth)

INPUT:

TrafficRate – frequency of data communication

MobilityRate – frequency of topology changes

DelayRequirement – acceptable route setup delay

Bandwidth – available bandwidth

OUTPUT:

Recommended routing approach

BEGIN

IF DelayRequirement = VERY_LOW AND TrafficRate = HIGH THEN

Recommendation ← PROACTIVE

ELSE IF TrafficRate = LOW AND Bandwidth = LIMITED THEN

Recommendation ← REACTIVE

ELSE IF MobilityRate = HIGH AND TrafficRate = INTERMITTENT THEN

Recommendation ← REACTIVE

ELSE IF TrafficRate = HIGH AND DelayRequirement = LOW THEN

Recommendation ← PROACTIVE

ELSE

Recommendation ← EVALUATE_BOTH

END IF

RETURN Recommendation

END

23. Route Selection and Metric Calculation

FUNCTION CalculateRouteMetric(HopCount, Delay, LossRate, EnergyCost)

```
INPUT:
HopCount, Delay, LossRate, EnergyCost
OUTPUT:
RouteMetric
BEGIN
RouteMetric ←
W1 × HopCount +
W2 × Delay +
W3 × LossRate +
W4 × EnergyCost
RETURN RouteMetric
END
ALGORITHM SelectBestRoute(RouteSet)
INPUT:
RouteSet – candidate routes
OUTPUT:
BestRoute
BEGIN
BestRoute ← NULL
BestMetric ← INFINITY
FOR each Route in RouteSet DO
Metric ← CalculateRouteMetric(Route.HopCount,
Route.Delay,
Route.LossRate,
Route.EnergyCost)
```

IF Metric < BestMetric THEN

BestMetric ← Metric

BestRoute ← Route

END IF

END FOR

RETURN BestRoute

END

24. Packet Forwarding in a Mobile Network

ALGORITHM ForwardMobilePacket(Node, Packet)

INPUT:

Node – current forwarding node

Packet – data packet

OUTPUT:

Delivery, forwarding or drop decision

BEGIN

Destination ← Packet.Destination

IF Node = Destination THEN

DELIVER Packet

RETURN DELIVERED

END IF

IF ValidRouteTo(Destination) THEN

NextHop ← RouteTable[Destination].NextHop

IF LinkStatus(NextHop) = UP THEN

FORWARD Packet TO NextHop

RETURN FORWARDED

ELSE

Mark route as INVALID

RETURN ROUTE_FAILURE

END IF

ELSE

RETURN ROUTE_DISCOVERY_REQUIRED

END IF

END

25. Routing Loop Detection

ALGORITHM DetectRoutingLoop(Path)

INPUT:

Path – ordered sequence of nodes

OUTPUT:

TRUE if loop exists, otherwise FALSE

BEGIN

Visited ← EMPTY_SET

FOR each Node in Path DO

IF Node IN Visited THEN

RETURN TRUE

END IF

ADD Node TO Visited

END FOR

RETURN FALSE

END

26. Manual Trace / Desk Check

Mobile routing pseudocode must be manually traceable. Record the node, current route state, received message, decision, next hop, and resulting route table state.

STEP	NODE	EVENT	DECISION	ROUTE STATE
1	S	Needs D	No valid route	UNKNOWN
2	S	Broadcast RREQ	Forward to neighbors	DISCOVERY
3	A	Receives RREQ	Record reverse route	DISCOVERY
4	D	Receives RREQ	Generate RREP	VALID
5	S	Receives RREP	Install route S→A→D	VALID
6	S	Data packet	Forward to A	ACTIVE

27. What Will NOT Be Accepted as Mobile Routing Pseudocode?

Too vague:

Start

Find route

Send packet

Handle movement

Display result

Stop

There is insufficient computational logic.

Programming code disguised as pseudocode:

for(int i=0; i<n; i++) {

if(route[i].valid) send(packet);

}

Acceptable algorithmic form:

FOR each Route in RouteTable DO

IF Route.State = VALID THEN

FORWARD Packet TO Route.NextHop

RETURN FORWARDED

END IF

END FOR

RETURN ROUTE_DISCOVERY_REQUIRED

28. Programming-Language Syntax Should Not Be Used

- Avoid Java, Python, C, C++, Android API calls, or simulator-specific commands unless actual implementation code is explicitly requested.
- Use SEND, RECEIVE, FORWARD, DELIVER, DROP, UPDATE, ADD ROUTE and REMOVE ROUTE.
- Use ← for assignment.
- Use explicit network conditions instead of vague statements such as 'handle mobility'.

29. Comments

// Detect whether the current link is still usable

IF LinkStatus(NextHop) = DOWN THEN

Mark Route INVALID

Initiate route repair

END IF

Comments may clarify a non-obvious networking decision, but comments cannot replace the routing logic. The supplied DAA standard similarly requires

the pseudocode itself to analyse how the result is obtained. filecite turn0file0 L271-L277

30. Quality Checklist

- Inputs are clearly defined.
- Outputs are clearly defined.
- Mobile-network states have clear meanings.
- Addresses, routes and sequence information are initialised.
- Packet forwarding conditions are explicit.
- Registration and handoff conditions are explicit.
- Route discovery and route repair have clear termination.
- Link failure and unavailable-route cases are addressed.
- The algorithm can be manually traced.
- Programming-language-specific syntax is avoided.
- Another programmer could implement the algorithm without guessing the logic.

31. How Mobile Computing Pseudocode Will Be Assessed

CRITERION	WHAT IS EXPECTED
Algorithmic Correctness	The mobility or routing logic correctly solves the stated problem.
Completeness	Inputs, outputs, state, decisions, loops, packet actions and termination are sufficiently specified.
Precision	Conditions such as route validity, link state, registration status and sequence number are explicit.
Networking Understanding	The solution correctly reflects Mobile IP, cellular/ad hoc networking or routing concepts.
Appropriate Abstraction	Network operations are represented clearly without unnecessary programming syntax.
Traceability	A small network scenario can be manually executed using the pseudocode.
Technical Justification	The selected proactive/reactive or mobility mechanism is supported by network characteristics.

32. Final Test

Before submitting any Mobile Computing pseudocode, ask:

- Can another competent programmer implement my algorithm without asking what I meant?
- Can the same algorithm be implemented in Java, Python, C/C++ or a network simulator without changing its underlying logic?
- Does the algorithm explicitly handle mobility, routing state and failure conditions?
- Can I trace the algorithm using a small mobile-network example?

33. THE MOBILE COMPUTING DESIGN RULE

UNDERSTAND → MODEL NETWORK → DEFINE STATE → DESIGN ALGORITHM → WRITE PSEUDOCODE → TRACE → TEST → REFINE → IMPLEMENT

Do not write the program first and convert it into pseudocode afterwards. Design the mobile-network behaviour first. The program or simulator is the implementation of that design. This follows the source handbook's final design principle:

THINK → DESIGN → WRITE PSEUDOCODE → TRACE → REFINE → IMPLEMENT.

ITA0327-MOBILE COMPUTING

ASSIGNMENT 3 – CO3

WIRELESS NETWORKS, MOBILE IP AND ROUTING FOR A MOBILE EMERGENCY COMMUNICATION NETWORK

Student details	Information
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Course	ITA0327 - Mobile Computing
Assignment	Assignment 3 – CO3
Submission Date	06/11/2026

Abstract

This report analyses wireless network architecture, Mobile IP, cellular versus ad hoc networking, Mobile IP applications, and proactive (DSDV) and reactive (AODV) routing, and then combines them into a design for a mobile emergency communication network. Worked examples show that handoff interruption of about 0.3 s costs 0.2% of packets at pedestrian speed (1.4 m/s) but 4.5% at 30 m/s, and that Mobile IP registration across a 40 ms one-way home path adds about 100 ms to each move. Tunnelling adds 20 bytes per packet, which is 2% for a 1,000-byte packet but 33% for a 60-byte voice packet. Triangular routing in the worked example stretches the path from 30 ms to 75 ms.

A small reproducible Python simulation (1,000 m × 1,000 m, 250 m radio range, random-waypoint mobility, 10 seeds per case) compares DSDV and AODV. At 40 nodes and 20 m/s, AODV delivered 85.0% of packets against 75.2% for DSDV, but AODV needed about 17,000 route-discovery messages against about 8,300 DSDV update messages. DSDV gave lower delay (23.0 ms against 26.9 ms). The final emergency design is a hybrid: a multi-hop ad hoc mesh among vehicles and rescue teams, AODV-style reactive routing, Mobile IP with Home Agent at the command centre for session continuity, and satellite or cellular gateways when they survive. The simulation is a simplified planning model, not a calibrated NS-2 or NS-3 run, and its limits are stated in Section Q7.

Pair-work declaration and Bloom's Taxonomy alignment

Student A is K. Goutham. Student B is the pair partner. Both students took part in every question; the primary responsibilities follow the assignment brief (the Contribution Record is at the end).

Bloom level	What this report demonstrates	Questions
Remember / Understand	Wireless network components, Mobile IP entities, cellular and ad hoc architecture, DSDV and AODV terminology	Q1–Q3, Q5, Q6
Apply	Handoff, delay and loss calculations; Mobile IP registration and tunnelling worked examples; DSDV table updates; AODV trace	Q1, Q2, Q5, Q6
Analyse	Triangular routing, tunnelling overhead, routing overhead against network size and mobility	Q2, Q5–Q7
Evaluate	Cellular versus ad hoc decision, DSDV versus AODV decision, suitability of Mobile IP	Q3, Q4, Q6, Q7
Create	Joint emergency network design, pseudocode, simulation workflow	Q7

Planning assumptions and selected parameters

Parameter	Adopted value	Reason
Wi-Fi AP radius	100 m, 20 m overlap	Typical outdoor planning value for the Q1 scenarios

Parameter	Adopted value	Reason
Layer-2 handoff interruption	0.3 s	Scan, authenticate and associate; typical order of magnitude, not a measured value
One-way delay MN–FA	5 ms	Single wireless hop
One-way delay FA–HA	40 ms	Wide-area path in the Q2 example
Per-hop delay in the simulation	10 ms	Planning value for ad hoc hops
Simulation area / radio range	1,000 m × 1,000 m / 250 m	Small incident area
Simulation duration	300 s per run	10 seeds averaged
Traffic	5 flows, 2 packets/s each	Light emergency messaging load
DSDV update policy	Full dump every 15 s plus triggered updates on neighbour change	Standard DSDV behaviour
AODV control	RREQ flood, RREP, RERR, plus 1 Hello/s per node	Standard AODV behaviour

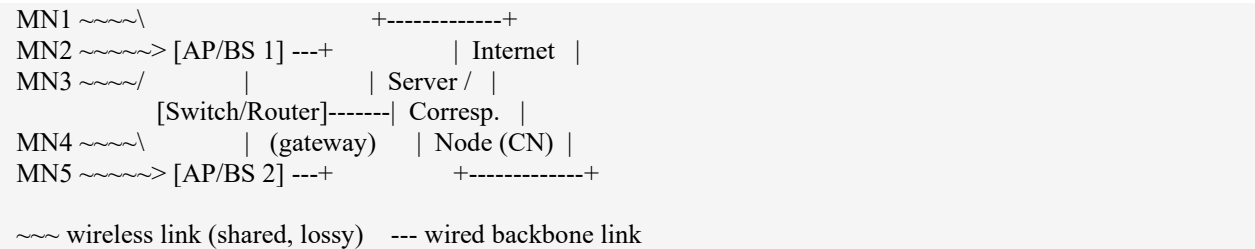
Modelling note: all values are planning assumptions. Real deployments need site surveys and measured traces.

Q1 – Wireless Network Architecture and Mobility Analysis (6 marks)

Part A – Introduction to wireless networks

Joint work. A wireless network connects mobile nodes (MNs) to each other and to a wired backbone through radio links. Its characteristics are: shared and lossy medium, limited bandwidth, variable link quality, node mobility, and limited battery power.

Student A – Network architecture analysis.



Component	Role
Mobile node (laptop, phone, vehicle unit)	Generates and receives traffic; selects and changes its point of attachment
Access point / base station	Provides the radio interface, allocates the channel,

Component	Role
	bridges wireless and wired segments
Switch / router / gateway	Forwards traffic between access points and external networks, applies addressing and policy
Wireless link	Carries data over a shared medium with interference and fading
Correspondent node / server	The peer or service the mobile node communicates with

Bandwidth effect, worked example: an AP with 54 Mbps nominal rate gives roughly 25 Mbps usable throughput after protocol overhead. Shared by 10 active users, each gets about 2.5 Mbps; with 50 users, about 0.5 Mbps. Connectivity effect: when signal strength falls below the threshold, throughput drops through rate adaptation, then the link breaks. Communication path: MN → radio link → AP → switch → gateway → CN, and the reverse for replies.

Student B – Mobility and performance analysis. Take an AP radius of 100 m, so a cell diameter of about 200 m. Dwell time in a cell is 200/v, the handoff interruption is 0.3 s, and the voice stream sends 50 packets/s.

Mobility condition	Speed	Dwell time per cell	Packets lost per handoff	Loss fraction (0.3 / dwell)
Pedestrian	1.4 m/s	143 s	15	0.21%
Vehicle	15 m/s	13.3 s	15	2.25%
Fast vehicle	30 m/s	6.7 s	15	4.5%

Changing conditions: as the number of users sharing an AP rises, queuing delay grows; as speed rises, handoff frequency doubles each time speed doubles. Packet loss therefore grows linearly with speed for a fixed interruption, so fast mobility needs faster handoff, not just better coverage.

Part B – Wireless mobility scenario

Joint scenario. A vehicle moves along a straight road at 15 m/s past three APs at x = 0 m, 180 m and 360 m (radius 100 m, overlap 20 m). It starts at x = 20 m.

Student A – Analytical investigation. Successful communication: MN attached and receiving packets within 40 ms round trip. Interrupted communication: the 0.3 s window while the MN scans and re-associates.

Time (s)	Position (m)	Event	State
0	20	Attached to AP1, strong signal	Successful
4.67	90	Signal in overlap zone falls below threshold, handoff starts	Interrupted
4.97	105	Associated with AP2, data resumes	Successful

Time (s)	Position (m)	Event	State
16.67	270	Overlap with AP3, handoff starts	Interrupted
16.97	275	Associated with AP3	Successful
28.0	440	Leaves AP3 coverage	Scenario ends

Total packets sent at 50 packets/s over 28 s = 1,400. Lost: 2 handoffs × 15 = 30, so packet delivery ratio = 1,370/1,400 = 97.9%. Increasing mobility: at 30 m/s the same road is crossed in 14 s with the same two handoffs, so the loss is still 30 packets but of only 700, giving 95.7%. A longer route with more APs makes the loss proportional to the number of handoffs.

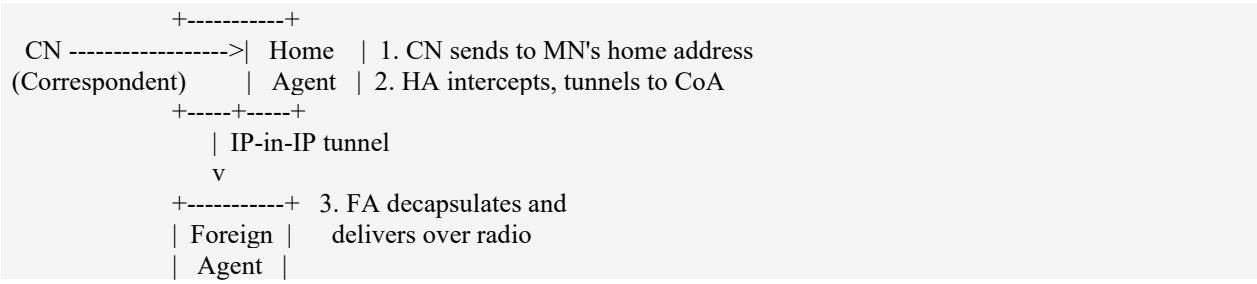
Student B – Mobility logic. The MN changes its point of attachment when the new AP's signal exceeds the current one by a hysteresis margin; otherwise the node would keep switching (ping-pong) at the cell boundary. The scenario terminates when no AP signal exceeds the minimum threshold or the route ends. Repeated mobility reduces efficiency because each handoff adds interruption and signalling.

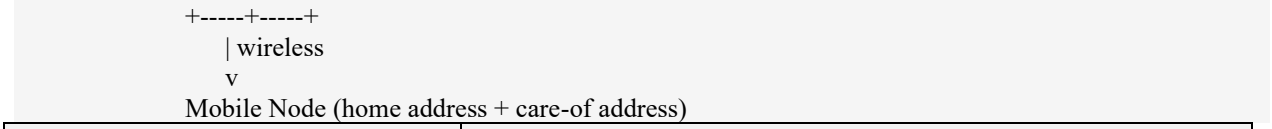
```
ALGORITHM SelectAttachment(MN, CurrentAP, VisibleAPs)
INPUT:  MN – mobile node; CurrentAP – serving AP; VisibleAPs – APs in range
OUTPUT: Attachment decision
BEGIN
 BestAP ← CurrentAP
 FOR each AP in VisibleAPs DO
  IF SignalStrength(AP) > SignalStrength(BestAP) + HYSTERESIS THEN
   BestAP ← AP
  END IF
 END FOR
 IF BestAP = CurrentAP THEN
  IF SignalStrength(CurrentAP) < MIN_SIGNAL THEN
   RETURN DISCONNECTED
  END IF
  RETURN STAY
 END IF
 PerformHandoff(MN, CurrentAP, BestAP)
 RETURN SWITCHED
END
```

Trace (15 m/s run): at t = 4.67 s, AP2 signal exceeds AP1 by more than the hysteresis margin, so the algorithm returns SWITCHED. At t = 10 s the MN is mid-cell, BestAP = CurrentAP and the signal is above MIN_SIGNAL, so it returns STAY.

Q2 – Mobile IP and Mobility Management (7 marks)

Architecture and terminology (joint)





Entity	Description
Mobile Node (MN)	Keeps a permanent home address while changing its point of attachment
Home Agent (HA)	Router on the home network; stores the binding (home address → CoA) and tunnels packets
Foreign Agent (FA)	Router on the visited network; advertises, offers the CoA and delivers decapsulated packets
Correspondent Node (CN)	The peer that communicates with the MN using its home address
Care-of Address (CoA)	Address that identifies the MN's current location

Student A – Mobile IP operation

1. **Agent discovery:** the MN listens for agent advertisements (or sends a solicitation) and learns whether it is at home or visiting.
2. **Registration / location update:** the MN sends a Registration Request (home address, CoA, lifetime, identifier) via the FA to the HA; the HA authenticates it, stores the binding and returns a Registration Reply.
3. **Packet delivery:** CN → home address → HA intercepts → encapsulates (outer header to CoA) → FA decapsulates → MN. Replies from the MN can go directly to the CN (triangular routing).

```
ALGORITHM RegisterMobileNode(MN, HA, CoA)
INPUT:  MN, HA, CoA
OUTPUT: Registration result
BEGIN
 Request.HomeAddress ← MN.HomeAddress
 Request.CareOfAddress ← CoA
 Request.Lifetime ← RequestedLifetime
 Request.Identifier ← GenerateRegistrationIdentifier()
 Authenticate Request
 SEND Request TO HA
 START TIMER RegTimeout
 WAIT FOR RegistrationReply
 IF Reply.Status = ACCEPTED THEN
  Binding[MN.HomeAddress] ← CoA
  RegistrationExpiry ← CurrentTime + Reply.Lifetime
  RETURN REGISTRATION_SUCCESS
 ELSE
  RETURN REGISTRATION_FAILURE
 END IF
END
```

Worked example (registration/packet-flow sequence).

Step	From → To	Action	Time (ms)

Step	From → To	Action	Time (ms)
1	FA → MN	Agent advertisement received	5
2	MN → FA	Registration Request	5
3	FA → HA	Request relayed	40
4	HA	Authenticate, update binding	10 (processing)
5	HA → FA → MN	Registration Reply	40 + 5
6	CN → HA → FA → MN	Data now tunnelled	–

Registration delay = 5 + 40 + 10 + 40 + 5 = 100 ms.

Student B – Mobile IP challenge analysis

Challenge	Analysis
Registration delay	100 ms in the example, plus the 0.3 s layer-2 handoff: about 0.4 s of total interruption per move
Tunnelling overhead	IP-in-IP adds 20 bytes: 20/1,000 = 2% for a 1,000-byte packet, but 20/60 = 33% for a 60-byte voice packet
Triangular routing	Forward path CN → HA (30 ms) → FA (40 ms) → MN (5 ms) = 75 ms, against a direct path of about 30 ms: stretch of 2.5 times
Packet loss during mobility	At 50 packets/s, 0.4 s of interruption loses about 20 packets unless the old FA or HA buffers them
Security	Registration messages must be authenticated, with replay protection through the identifier, otherwise an attacker could redirect traffic
Scalability	Every move sends a registration to a possibly distant HA; many mobile nodes concentrate load on one HA

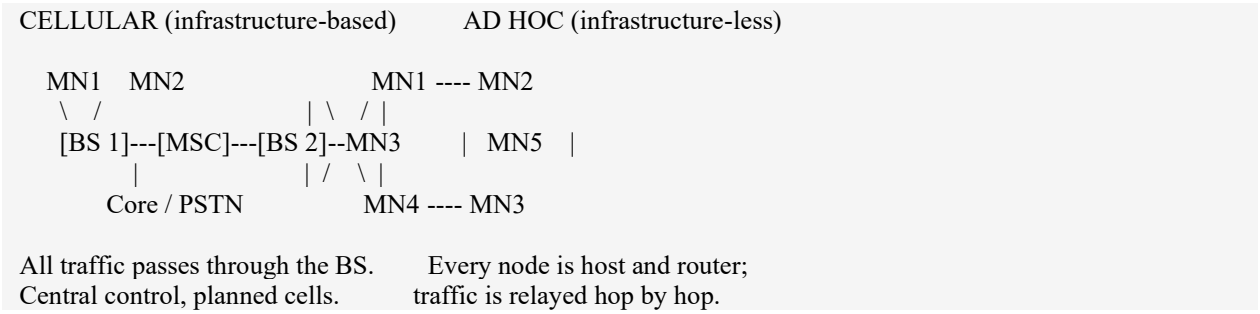
Joint design question

If a mobile user changes its network of attachment frequently during a session, three problems occur. First, each move costs about 0.4 s of interruption, so at one move every 5 s the node is unreachable about 8% of the time. Second, registration signalling to the distant HA grows linearly with move rate. Third, packets sent to the old CoA before the new binding is installed are lost. They can be reduced by: **regional or hierarchical registration** so that local moves are registered with a nearby gateway instead of the HA; **route optimisation** (binding updates to the CN) to remove triangular routing; **buffering and forwarding by the previous agent** (smooth handoff) to recover in-flight packets; **fast agent discovery** with link-layer triggers so that registration begins before the old link breaks; and **longer registration lifetimes** with authenticated renewals to cut periodic signalling. Justification: the first and fourth items attack the 0.4 s interruption, the second removes the 2.5 times path stretch, and the third removes most of the loss during a move.

Cross-understanding. Student A can state these challenges; Student B can describe the full registration and tunnelling flow above.

Q3 – Cellular Networks versus Ad Hoc Wireless Networks (7 marks)

Architectures (joint)



Student A – Cellular network investigation

The service area is divided into hexagonal cells, each served by a base station (BS) connected to a mobile switching centre (MSC) and the core network. The BS allocates channels and the MSC controls handoff, authentication and billing. Communication is **centralised**: even two phones in the same cell communicate via the BS. **Handoff** between cells is controlled by the network using signal-strength measurements, so a call continues while the user moves. **Scalability** comes from frequency reuse and cell splitting (Assignment 1 showed that a 19-cell layout carrying about 729 Erlangs scales to 76 cells and about 2,917 Erlangs after one uniform split). **Coverage** is planned and wide. Suitable applications: public voice and mobile data, nationwide coverage, and billed services.

Why cellular networks depend on infrastructure: channels, addressing, authentication and handoff decisions all sit in the BS and MSC. If the BS loses power or its backhaul fails, every mobile in that cell loses service even though the phones themselves are working.

Student B – Ad hoc network investigation

An ad hoc network is formed by mobile nodes alone, with no fixed BS. Each node acts as both host and router. Communication between nodes that are out of direct range is **multi-hop**: intermediate nodes forward the packets. The **topology is dynamic** because nodes move, switch off or run out of battery, so routes must be discovered and repaired (DSDV and AODV, Q5–Q6). There is **no fixed infrastructure**, so the network can be deployed in minutes. Suitable applications: disaster recovery, military operations, vehicular networks, temporary events and field work.

How ad hoc networks continue without fixed infrastructure: each node keeps a neighbour list and routing state and forwards for others; when one relay fails, the routing protocol finds another path through remaining nodes, provided the network is still connected.

Comparative analysis

Parameter	Cellular network	Ad hoc network
Infrastructure	BS, MSC and backhaul required	None required

Parameter	Cellular network	Ad hoc network
Communication	Centralised through the BS	Direct or multi-hop peer to peer
Routing	Infrastructure-assisted; static backbone routes	Distributed route discovery and maintenance
Mobility	Managed by the network (location registers, handoff)	Handled by every node through route repair
Topology	Planned cells, stable	Dynamic and unpredictable
Scalability	Good: add cells, reuse frequencies	Limited by control overhead and capacity falling with hop count
Handoff	Network-controlled, hard or soft	No handoff; routes are re-established
Failure handling	A BS failure disconnects its cell	Alternative peer routes if the network stays connected
Applications	Public mobile services	Emergency, military, temporary and vehicular networks

Joint engineering decision

For an emergency or disaster area where fixed infrastructure may be unavailable, we select an **ad hoc (multi-hop) architecture**, backed by a gateway to any surviving cellular or satellite link. Reasons: it can be deployed immediately (the decision rule in the SIMATS pseudocode standard selects AD_HOC when infrastructure is UNAVAILABLE and deployment time is IMMEDIATE); no single device is a critical point of failure; and every vehicle and handset extends coverage. The trade-off is lower and less predictable capacity, which is why we keep the gateway option.

```
ALGORITHM SelectWirelessArchitecture(Scenario)
INPUT:  InfrastructureAvailability, DeploymentTime, CoverageRequirement, MultiHopRequirement
OUTPUT: Recommended architecture
BEGIN
  IF InfrastructureAvailability = AVAILABLE AND CoverageRequirement = LARGE THEN
    Architecture ← CELLULAR
  ELSE IF DeploymentTime = IMMEDIATE AND InfrastructureAvailability = UNAVAILABLE THEN
    Architecture ← AD_HOC
  ELSE IF MultiHopRequirement = REQUIRED THEN
    Architecture ← AD_HOC
  ELSE
    Architecture ← CELLULAR
  END IF
  RETURN Architecture
END
```

Trace for the disaster case: InfrastructureAvailability = UNAVAILABLE and DeploymentTime = IMMEDIATE, so the first condition fails and the second succeeds; the result is AD_HOC.

Q4 – Applications of Mobile IP and Routing Requirements (7 marks)

Selected application: mobile healthcare (ambulance telemedicine)

Using the SelectMobileIPApplication rule, a requirement for remote patient connectivity returns MOBILE_HEALTHCARE. An ambulance carries a patient and sends vital signs and live voice to a hospital specialist while driving across several networks.

Student B – Application architecture

Element	Role
Mobile node	Ambulance communication unit, with a permanent home address registered at the hospital network
Home Agent	Router at the hospital; keeps the binding home address → current CoA and tunnels packets
Foreign Agents	Gateways of the cellular or Wi-Fi networks along the route; give the CoA
Correspondent node	Specialist workstation and patient-record server at the hospital
Communication endpoints	Ambulance unit and hospital server

How Mobile IP supports mobility: the hospital server always addresses the ambulance by its home address. When the ambulance enters a new network it registers a new CoA with the HA and the session continues without the application noticing the address change.

Communication flow: ambulance leaves network 1 → agent discovery in network 2 → registration with the HA (about 100 ms in the Q2 example) → HA tunnels hospital packets to the new CoA → ambulance replies reach the hospital directly or through a reverse tunnel.

Student A – Performance and requirement analysis

Requirement	Analysis
Communication delay	Tunnelled one-way path 75 ms in the Q2 example, within the 150 ms one-way guideline commonly used for interactive voice, but with little margin if the HA is far away
Packet delivery / connectivity	Vital signs need at least 99% delivery; sent over TCP-like reliable transport or with retransmission
Interruption during movement	About 0.4 s per move (0.3 s layer-2 handoff plus 0.1 s registration). With 1 km cells at 15 m/s, dwell is 67 s, so 0.6% downtime; with 200 m cells, 13.3 s dwell, so 3% downtime
Mobility requirements	Fast agent discovery, buffering during handoff, and smaller registration delay; regional registration if moves are

Requirement	Analysis
	frequent
Security	Authenticated registration, an IPsec tunnel or equivalent for patient data, and protection against replay
Reliability	Two radio interfaces (for example cellular plus Wi-Fi) so one link can bridge a handoff

Routing requirements created by mobility: the HA must update its binding quickly; the route to the CoA must be re-evaluated after each move; and the triangular path should be shortened by route optimisation for long sessions.

Evaluation of suitability. Mobile IP is suitable because the session must survive address changes, and the hospital-centred architecture gives a natural HA. It is less suitable if moves are very frequent (3% downtime at 200 m cells is noticeable for video), which is why we add buffering and regional registration. Without these additions, the delay and interruption are acceptable for vital-sign telemetry but borderline for live video.

Q5 – Proactive Routing Protocols in Mobile Networks (6 marks)

Joint study

A proactive protocol maintains routes to every destination before data is sent, by periodic and triggered exchange of routing tables. This gives low route-setup delay at the cost of continuous control traffic. We select **DSDV** (Destination-Sequenced Distance Vector), a distance-vector protocol that adds destination sequence numbers to prevent routing loops.

Student A – Protocol design

Principle: every node keeps a table with an entry for every destination: next hop, hop count (metric) and destination sequence number. Nodes advertise their tables to neighbours periodically, and immediately when a link changes (triggered update). Sequence numbers: each destination increments its own even sequence number on every advertisement; a node that detects a broken link advertises the lost destination with an odd sequence number and an infinite metric. A received route replaces the stored one if its sequence number is higher, or if it has the same sequence number and a lower hop count.

```
ALGORITHM UpdateDSDVRoute(Node, ReceivedRoute)
INPUT:  Node, ReceivedRoute = {Destination, NextHop, HopCount, SequenceNumber}
OUTPUT: Updated routing table
BEGIN
 Destination ← ReceivedRoute.Destination
 NewHopCount ← ReceivedRoute.HopCount + 1
 NewSequence ← ReceivedRoute.SequenceNumber
 IF Destination not in RouteTable THEN
   ADD ROUTE Destination via sender with NewHopCount, NewSequence
 ELSE IF NewSequence > RouteTable[Destination].SequenceNumber THEN
   UPDATE ROUTE Destination via sender with NewHopCount, NewSequence
 ELSE IF NewSequence = RouteTable[Destination].SequenceNumber AND
        NewHopCount < RouteTable[Destination].HopCount THEN
   UPDATE ROUTE Destination via sender with NewHopCount, NewSequence
 END IF
```

```
  RETURN RouteTable
END
```

Sample topology. Nodes A, B, C, D, E with links A–B, B–C, C–D, B–E, E–D.

```
    A ---- B ---- C
           |      |
           E ---- D
```

Current sequence numbers: A = 100, B = 200, C = 300, D = 400, E = 500.

Routing table of A before the topology change

Destination	Next hop	Metric	Sequence
A	A	0	100
B	B	1	200
C	B	2	300
D	B	3	400
E	B	2	500

Topology change: link C–D fails. C marks D unreachable and advertises D with metric ∞ and odd sequence 401. B and A accept it because 401 > 400. An older advertisement from E (D, metric 1, sequence 400) is now ignored because 400 < 401. D then advertises itself with the next even sequence 402; E accepts D at metric 1, B accepts D via E at metric 2, and A accepts D via B at metric 3.

Routing table of A after the change

Destination	Next hop	Metric	Sequence
A	A	0	100
B	B	1	200
C	B	2	300
D	B (now through E)	3	402
E	B	2	500

The next hop and metric for D look the same, but the sequence number has moved from 400 to 402, which shows the route is fresh and now goes through E instead of C. A packet from A to D follows A → B → E → D after convergence.

Student B – Performance investigation

Routing-control overhead. With N nodes, each full dump carries N entries. Assuming 12 bytes per entry plus 8 bytes of header, one dump is 12N + 8 bytes, and the whole network sends N dumps per 15 s period.

Nodes N	Bytes per dump	Periodic control traffic (bytes/s, whole network)
10	128	10 × 128 / 15 = 85
20	248	20 × 248 / 15 = 331

Nodes N	Bytes per dump	Periodic control traffic (bytes/s, whole network)
40	488	40 × 488 / 15 = 1,301

Doubling N roughly quadruples periodic overhead (growth is proportional to N squared). Triggered updates add more when the topology is unstable.

Route availability and convergence. A route is available as soon as the table has a valid entry, so there is no discovery delay. After a link break, convergence takes about (number of hops to propagate) × per-hop delay: in the example 3 hops × 10 ms = 30 ms for triggered updates, but a node that has to wait for the destination's next even-sequence advertisement may wait up to one update period (15 s), which is why stale routes are possible in DSDV.

Delay and delivery. Data packets use the stored next hop, so delay is hop count times 10 ms (about 17 ms to 24 ms average in the Q7 simulation). Delivery falls when routes break faster than updates arrive; the Q7 simulation records this against network size and speed.

Q6 – Reactive Routing and Protocol Comparison (7 marks)

Student A – Reactive routing investigation

Concept. A reactive protocol finds a route only when a source has data to send and no valid route exists. Nothing is maintained for idle destinations, so control traffic is demand-driven. **AODV** (Ad hoc On-demand Distance Vector) keeps one table entry per active destination: next hop, hop count, destination sequence number, state and lifetime.

Message	Purpose
Route Request (RREQ)	Broadcast by the source: source, destination, request ID, hop count, sequence numbers. Each node that receives it for the first time records a reverse route to the source and rebroadcasts it
Route Reply (RREP)	Unicast back along the reverse path by the destination (or a node with a fresh route); each node on the way installs the forward route
Route Error (RERR)	Sent when a link on an active route breaks; tells the affected sources which destinations are no longer reachable
Hello	Optional 1 s beacon to detect neighbour loss

Route discovery: source checks its table; if there is no valid route it increments the request ID, broadcasts a RREQ and waits; duplicates (same source and request ID) are discarded. **Route maintenance:** active routes are monitored; on link failure the node marks the route INVALID, tries local repair or sends RERR; the source then starts a new discovery if it still has data. **Effect of mobility:** every broken active route causes a RERR and a new RREQ flood, so discovery frequency (and delay) rises with mobility, but only for routes that are actually in use.

```
ALGORITHM ProcessRouteRequest(Node, Request)
INPUT:  Node, Request = {Source, Destination, RequestID, HopCount}
OUTPUT: Route discovery action
BEGIN
  IF Request already processed THEN
    DROP Request
```

```
  RETURN
 END IF
 ADD ROUTE toward Request.Source (reverse route) via sender
 Mark Request as processed
 IF Node = Request.Destination THEN
  SEND RouteReply toward Request.Source
  RETURN
 END IF
 IF ValidRouteTo(Request.Destination) THEN
  SEND RouteReply toward Request.Source using the valid route
 ELSE
  Request.HopCount ← Request.HopCount + 1
  FOR each Neighbor in NeighborTable DO
   FORWARD Request TO Neighbor
  END FOR
 END IF
END
```

Manual trace. Topology: S–A, A–D, S–B, B–C, C–D. S wants to send to D.

Step	Node	Event	Decision	Route state
1	S	Needs D	No valid route	UNKNOWN
2	S	Broadcast RREQ (ID 1)	Forward to A and B	DISCOVERY
3	A, B	Receive RREQ	Record reverse route to S; A forwards to D, B forwards to C	DISCOVERY
4	D	Receives RREQ via A (hop count 2)	Generate RREP; later copy via C is a duplicate and is dropped	VALID
5	S	Receives RREP via A	Install route S → A → D (2 hops)	VALID
6	S	Data packet	Forward to A	ACTIVE
7	A	Link A–D fails	Mark INVALID, send RERR to S	INVALID
8	S	RERR received, still has data	New RREQ (ID 2); route found S → B → C → D (3 hops)	VALID

Student B – Comparative verification: AODV and DSDV

Aspect	DSDV (proactive)	AODV (reactive)
Route discovery	None needed; routes are already in the table	RREQ flood and RREP when a route is needed
Routing table	Entry for every destination (size proportional to N)	Entries only for active destinations
Control overhead	Periodic and triggered updates, growing about as N squared for full dumps	Discovery floods per route break plus Hello beacons; grows with traffic and mobility

Aspect	DSDV (proactive)	AODV (reactive)
Initial communication delay	Low (no setup)	Setup delay about 2 × hops × per-hop delay, 60 ms for a 3-hop route at 10 ms per hop
Mobility handling	Triggered updates and sequence numbers; stale routes possible until next update	RERR and rediscovery for broken active routes only
Suitable scenarios	Small or medium networks, frequent traffic to many destinations, strict delay	Large or sparse networks, intermittent traffic, limited bandwidth, high mobility

```
ALGORITHM SelectRoutingApproach(TrafficRate, MobilityRate, DelayRequirement, Bandwidth)
INPUT:  TrafficRate, MobilityRate, DelayRequirement, Bandwidth
OUTPUT: Recommended routing approach
BEGIN
  IF DelayRequirement = VERY_LOW AND TrafficRate = HIGH THEN
    Recommendation ← PROACTIVE
  ELSE IF TrafficRate = LOW AND Bandwidth = LIMITED THEN
    Recommendation ← REACTIVE
  ELSE IF MobilityRate = HIGH AND TrafficRate = INTERMITTENT THEN
    Recommendation ← REACTIVE
  ELSE IF TrafficRate = HIGH AND DelayRequirement = LOW THEN
    Recommendation ← PROACTIVE
  ELSE
    Recommendation ← EVALUATE_BOTH
  END IF
  RETURN Recommendation
END
```

Joint question: DSDV or AODV for a highly dynamic network?

We select **AODV**. Using the simulation in Q7 (40 nodes, 20 m/s), AODV delivered 85.0% of packets against 75.2% for DSDV, and its control traffic was about 636 KB against about 1,166 KB for DSDV over 300 s, because it floods only for routes in use while DSDV re-advertises tables after every topology change. DSDV had slightly lower delay (23.0 ms against 26.9 ms) because no route setup is needed. Overhead reasoning: DSDV overhead grows with network size and with the rate of topology change, even when no one is sending; AODV overhead grows with the number of active flows and route breaks. Delay reasoning: AODV pays about 60 ms setup per discovery, which is small for messaging and acceptable for voice. Mobility reasoning: DSDV routes go stale between updates. Limits: in the small 10-node case DSDV used fewer control bytes at low speed (33 KB against 76 KB at 2 m/s), and with many simultaneous flows AODV's advantage shrinks, so the decision is for the highly dynamic, lightly loaded emergency network and should be re-evaluated for heavy traffic.

Q7 – Design a Mobile Emergency Communication Network (10 marks)

Scenario. After a natural disaster, ambulances, police vehicles, rescue teams and medical staff move across several areas; fixed infrastructure may be down; critical information must reach its destination with minimum delay.

Stage 1 – Independent design proposals (retained)

Student A – initial proposal

Item	Proposal
Wireless architecture	Hybrid: ad hoc multi-hop mesh between vehicles and teams, with gateway vehicles that connect to any working cellular tower or a satellite terminal
Mobility management	Location registration at a command-centre home agent; units change care-of address when they join another gateway's mesh
Role of Mobile IP	Keeps each unit's address constant so sessions with the command centre survive movement between gateway areas
Communication model	Priority unicast for voice and medical data, group broadcast for alerts

Student B – initial proposal

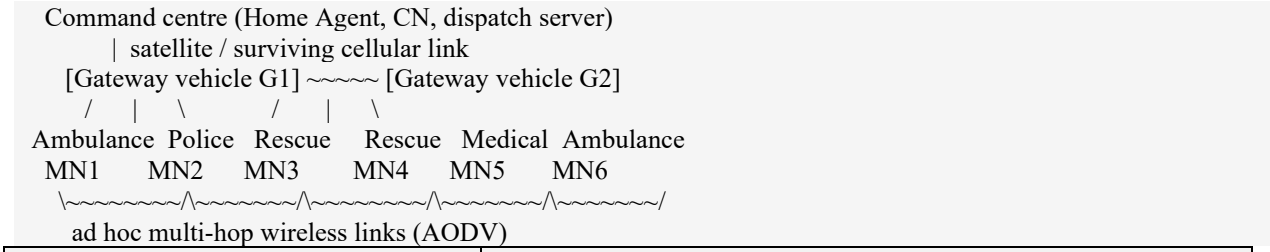
Item	Proposal
Routing strategy	Distributed multi-hop routing without any central router
Protocol	Reactive AODV, because traffic is bursty and topology changes often; DSDV only inside a stationary command post
Route maintenance	Hello beacons at 1 s, link-layer failure detection, RERR to sources, local repair before rediscovery, short route lifetimes
Failure handling	Alternate route, then buffer and retry; if the network is partitioned, store packets until a relay returns
Performance parameters	Packet delivery ratio, end-to-end delay, control overhead, route discovery time, hop count, energy per node

Stage 2 – Comparison of the two designs

Aspect	Where they agree	Where they differ	Strength	Weakness
Architecture	Multi-hop ad hoc, no dependence on towers	A adds gateways; B concentrates on routing inside the mesh	Resilient, quick to deploy	Capacity falls with hop count
Mobile IP	Needed for sessions with the command centre	A requires it everywhere; B uses it only at the gateway boundary	Keeps addresses stable	Registration delay and tunnel overhead
Routing suitability	Reactive routing suits bursty, mobile traffic	B gives detailed repair rules; A leaves routing unspecified	Low idle overhead	Discovery delay

Aspect	Where they agree	Where they differ	Strength	Weakness
Expected overhead	Both expect some signalling	A: registration and tunnel; B: RREQ floods and Hellos	–	Floods grow with node count
Expected delay	Both want minimum delay	A: 100 ms registration; B: about 60 ms discovery	–	Sum is about 160 ms after a gateway change
Node mobility	Both expect continuous movement	A handles movement between gateways; B handles movement within the mesh	Covers both levels	Needs coordination

Stage 3 – Joint final design



Component	Design decision
Mobile nodes	Ambulances, police vehicles, rescue teams, medical staff; each has a fixed home address at the command centre
Wireless links	Short-range radio (for example Wi-Fi mesh) between units; gateway links by satellite or cellular where available
Home Agent	At the command centre; stores the binding home address → current CoA
Foreign Agents	Gateway vehicles; each advertises its mesh and provides the CoA
Correspondent node	Dispatch server and hospital systems
Registration and tunnelling	A unit that joins a new gateway's mesh registers (about 100 ms), the HA tunnels packets to the gateway with 20 bytes of overhead, and the gateway delivers them over the mesh
Routing protocol	AODV inside each mesh with local repair; justification from Q6 and the test results below
Failure handling	Node or link failure: RERR, local repair, rediscovery. Gateway loss: re-register through another gateway. Network disconnection: buffer messages, retry on a timer, deliver when connectivity returns, and keep local unit-to-unit messaging

Component	Design decision
	working
Performance evaluation	Delay, packet delivery and routing overhead from the experiments below

```
ALGORITHM HandleLinkFailure(Node, Route, Packet)
INPUT:  Node, failed Route, Packet awaiting delivery
OUTPUT: Recovery status
BEGIN
  Mark Route INVALID
  IF AlternateValidRoute exists THEN
    FORWARD Packet via alternate route
    RETURN ALTERNATE_ROUTE_USED
  END IF
  SEND RouteError toward Source
  Initiate RouteDiscovery with RequestID + 1
  START TIMER DiscoveryTimeout
  IF RouteReply received before DiscoveryTimeout THEN
    FORWARD Packet via new route
    RETURN ROUTE_RECOVERED
  ELSE
    BUFFER Packet
    START TIMER RetryTimer
    RETURN DISCONNECTED_BUFFERED
  END IF
END
```

Stage 4 – Validation

Student A – Analytical validation.

Quantity	Calculation	Result
Mobile IP signalling per move	Request plus reply, about 100 bytes each	About 200 bytes per move
Signalling load, 40 units changing gateway every 60 s	40 × 200 / 60	About 133 bytes/s
Tunnelling overhead	20 / 1,000 for data; 20 / 60 for voice	2% and 33%
Mobility-related delay per gateway change	Registration 100 ms plus AODV rediscovery 2 × 3 hops × 10 ms = 60 ms	About 160 ms
Communication interruption	160 ms per change, one change per 60 s	About 0.27% downtime
Packet delivery	From simulation (below)	75% to 97% depending on density and speed

Justification: registration at the gateway boundary only (not for every mesh hop change) keeps signalling low, and AODV handles movement inside the mesh without involving the home agent.

Student B – Experimental validation. A small Python simulation was written for this report (file mc_sim.py, provided with the submission). Setup: 1,000 m × 1,000 m area, 250 m radio range, random-

waypoint movement without pause, 300 s per run, 5 flows at 2 packets/s, 10 random seeds averaged, 10 ms per hop. DSDV: full dump every 15 s plus a triggered update whenever a node's neighbour set changes; update size 12 bytes per entry plus 8 bytes. AODV: RREQ flood (24 bytes), RREP and RERR (20 bytes), Hello at 1 per second per node (20 bytes), a 5 s back-off after a failed discovery. Results below are outputs of that program, averaged over seeds; they are model results, not field measurements.

Nodes	Speed (m/s)	PDR DSDV / AODV (%)	Delay DSDV / AODV (ms)	Control traffic DSDV / AODV (KB per 300 s)	AODV route discoveries
10	2	49.7 / 49.6	16.6 / 19.9	33 / 76	160
10	10	40.8 / 40.1	17.0 / 19.8	60 / 87	223
10	20	37.8 / 36.7	16.8 / 20.0	88 / 96	267
20	2	80.1 / 80.3	24.4 / 29.0	133 / 150	88
20	10	74.1 / 75.3	23.3 / 27.2	239 / 207	196
20	20	64.8 / 68.2	23.5 / 28.3	343 / 280	337
40	2	96.4 / 97.2	23.5 / 26.9	513 / 279	40
40	10	85.6 / 90.7	23.5 / 26.8	891 / 452	206
40	20	75.2 / 85.0	23.0 / 26.9	1,166 / 636	380

Interpretation. (1) With only 10 nodes in the area the network is often partitioned, so delivery is about 40% to 50% for both protocols: no routing protocol can deliver when no path exists, which is why the design includes buffering and gateways. (2) Delivery rises with density (96% to 97% at 40 nodes and low speed) and falls with speed. (3) DSDV is faster by about 3 to 6 ms because it needs no discovery. (4) AODV's advantage in delivery and control bytes grows with node count and speed; at 10 nodes DSDV is cheaper. (5) Both proactive and reactive behaviour match the Q6 table.

Limits of the model: simplified radio (fixed range, no interference or collisions), no MAC layer, one-step lag for DSDV convergence, light traffic, and no energy model. A final evaluation should repeat these tests in NS-3 or on a testbed.

Stage 5 – Cross-defence

Student A – why the selected routing protocol was appropriate. AODV is appropriate because emergency traffic is bursty and the topology changes continuously; AODV keeps routes only for active flows, so idle units cost little overhead; it repairs breaks locally; and in the dense, fast cases it delivered more packets with fewer control bytes than DSDV.

Student B – how Mobile IP maintains communication when a node changes location. The unit keeps its home address. After moving, it discovers the new gateway, obtains a CoA and registers it with the HA, which updates the binding. The command centre keeps sending to the home address; the HA intercepts, tunnels to the CoA, and the gateway decapsulates and delivers. Registration takes about 100 ms, during which packets may be buffered or lost.

Stage 6 – Joint evaluation

Question	Answer
Why the architecture is suitable	Multi-hop ad hoc mesh needs no fixed infrastructure, deploys immediately and survives single failures; gateways give a path to command and hospitals
Why Mobile IP is needed	Sessions with the command centre must keep working when a unit moves between gateway areas; it is not needed for purely local unit-to-unit communication inside one mesh
Why the routing protocol is appropriate	AODV suits bursty traffic and high mobility (Q6, Q7 results)
Overhead introduced	Registration (about 200 bytes per move), 20 bytes per tunnelled packet, RREQ floods and Hellos (a few hundred KB per 300 s for 40 nodes in the model)
How mobility affects performance	Delivery fell from 97.2% to 85.0% (AODV, 40 nodes) as speed rose from 2 to 20 m/s; each gateway change costs about 160 ms
When connectivity is lost	Messages are buffered, retried and delivered when a path returns; local messaging continues inside each partition
When the solution is not appropriate	Very heavy continuous traffic (such as many video streams), very sparse deployments with no relays, or hard real-time requirements below the 160 ms gateway-change delay

Conclusion. Before choosing Mobile IP and a routing protocol for a highly mobile wireless environment, a network designer should analyse: node mobility rate and speed; node density and connectivity (partition risk); traffic pattern, volume and delay requirement; available bandwidth and energy; infrastructure availability and gateway options; whether sessions must survive changes of network; security needs; and scalability as nodes increase. In this design those characteristics point to an ad hoc mesh with AODV routing and Mobile IP at the gateway boundary.